

PROBLEM AND WHO HAS IT

One rain gauge decides the payout. Nobody checks the gauge.

Index crop insurance pays when the station records too little rain. The claims officers at microinsurers and cooperatives settle on that number.

6.1 mm

Lagos gauge, July 2025: 10 of 31 days reported

131.7 mm

Satellite over the farm: 31 of 31 days

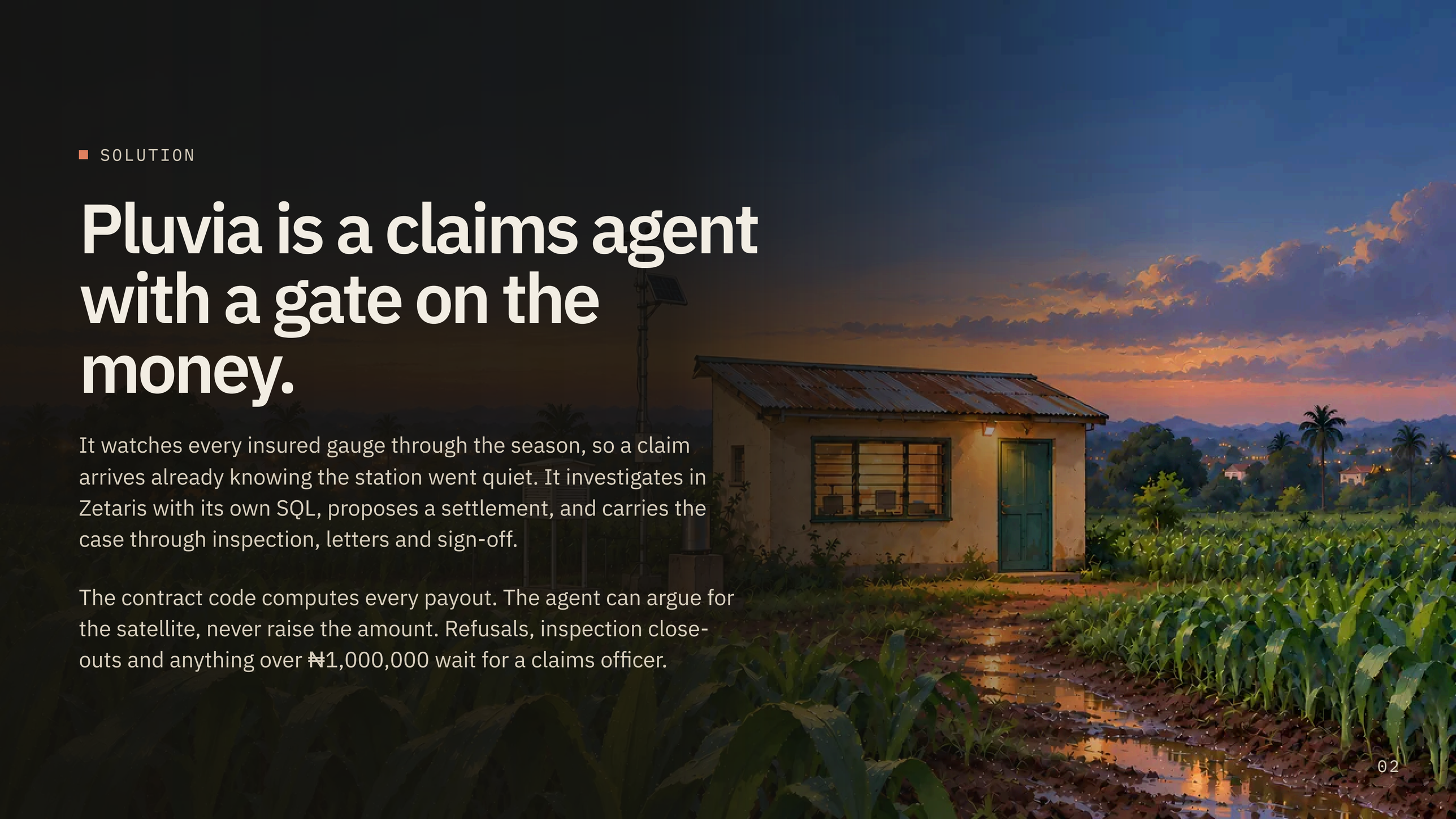
The automatic trigger reads the silence as drought and pays all three Lagos groups in full.

SOLUTION

Pluvia is a claims agent with a gate on the money.

It watches every insured gauge through the season, so a claim arrives already knowing the station went quiet. It investigates in Zetaris with its own SQL, proposes a settlement, and carries the case through inspection, letters and sign-off.

The contract code computes every payout. The agent can argue for the satellite, never raise the amount. Refusals, inspection close-outs and anything over ₦1,000,000 wait for a claims officer.



AGENT WORKFLOW

It plans, investigates, argues, and takes questions mid-case.

Watch

Checks each insured gauge daily. Opens a watch case on the third silent day.

Plan

Writes its steps before it touches the data, then ticks them off. A change of course needs a reason.

Investigate

Lists the Zetaris views and writes its own SQL: gauge against satellite, day by day, and 24 earlier Julys.

Propose

Cites each figure with the SQL behind it.

Gate

Code reruns every cited query and computes the payout from the contract.

Act

Sends an inspector, writes to the farmers and the station operator, asks for sign-off.

THE OFFICER ASKS

“Why did you not trust the gauge on 8 July?” It checks with its tools, answers with the figures, then picks its plan back up.



TECHNICAL STACK

One TypeScript service, live or replayed.

Reasoning

NVIDIA Nemotron 3 Ultra 550B with native tool calls, through Meterless Nightshift. GPT-5.6 Luna as fallback.

Data

Zetaris views over NOAA GHCN station records on public S3 and NASA POWER satellite rainfall over REST.

Memory

Meterless H-MEM keeps station lessons and precedents across cases. 13 of 13 on official conformance.

Service

Node and Hono, Zod at every boundary, a SQL guard that allows one SELECT over Pluvia views.

Desk

Plain HTML, CSS and JavaScript with Chart.js. Self-hosted fonts, no tracking.

Proof

129 tests. Every live case is saved with each query and model step, and replays offline.

HOW ZETARIS AND METERLESS ARE USED

Zetaris is where the agent works.

The agent discovers the tables and reads their columns. It writes every query itself. The watcher reads the same views once a simulated day, and the gate reruns the agent's own SQL before it accepts a number.

Nightshift serves the agent's steps on Nemotron, tools passed straight through. H-MEM hands the agent what earlier cases learned about the station.

Its first try at 24 earlier Julys failed in Zetaris. It read the error and wrote this.

```
SELECT
  CAST(SUBSTR(CAST(obs_date AS STRING), 1, 4) AS INT) AS year,
  COUNT(*) AS days,
  SUM(mm) AS total_mm,
  AVG(mm) AS avg_mm
FROM pluvia.power_lagos_2025_prior_daily
WHERE latitude = 6.577 AND longitude = 3.321
  AND CAST(SUBSTR(CAST(obs_date AS STRING), 6, 2) AS INT) = 7
  AND CAST(SUBSTR(CAST(obs_date AS STRING), 9, 2) AS INT) BETWEEN 1 AND 31
GROUP BY CAST(SUBSTR(CAST(obs_date AS STRING), 1, 4) AS INT)
ORDER BY year
LIMIT 2000
```

WHAT IS UNIQUE ABOUT THE SPONSOR USE

The model argues. The code checks its working.

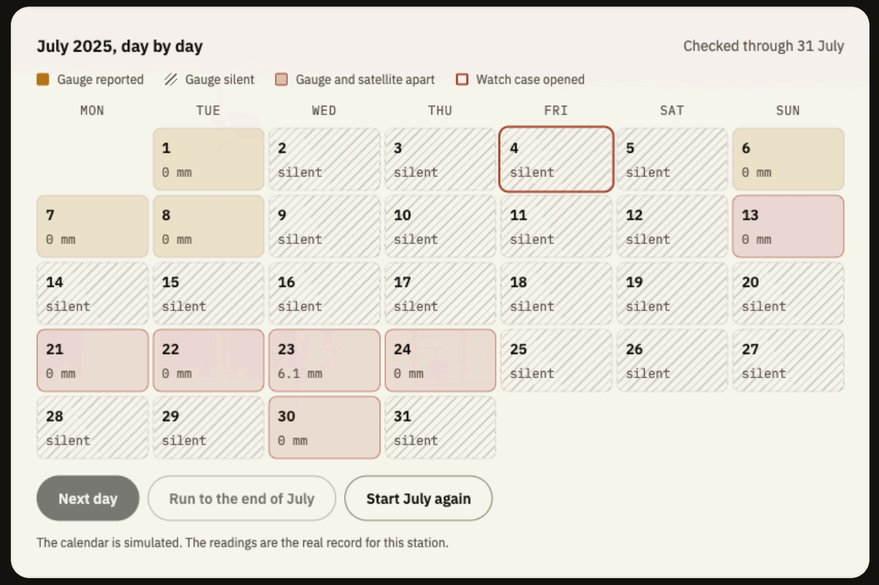
When Nemotron argues that the satellite should settle a claim, every number it cites travels with its SQL. Pluvia reruns that SQL in Zetaris and compares to 0.1 mm. The satellite only wins when its July sits inside its own range for 24 earlier Julys at the same spot.

On claim 001 it cited five figures. The code reran all five in Zetaris and every one held. July's 131.65 mm sat inside 117.06 to 416.37 mm, so the satellite settled it: ₦383,333.33, released.



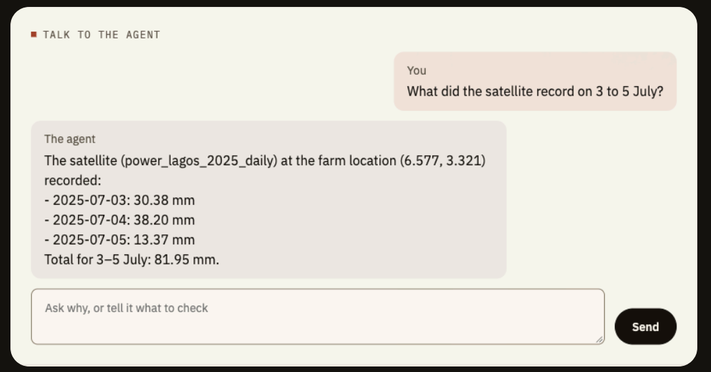
PRODUCT

The station, the case room, the decision.

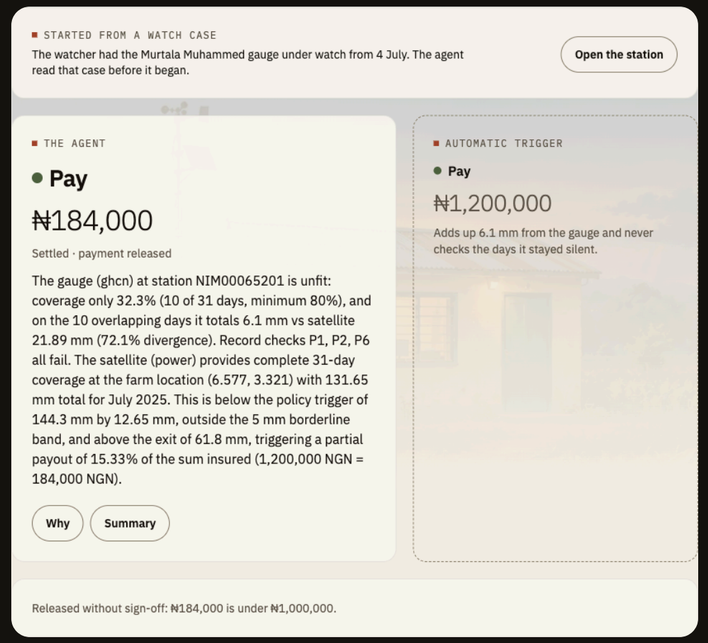


The watcher opens a case on 4 July, before any claim.





Its plan, ticked off, and the officer's question answered from its own query.



What the record supports, beside the automatic trigger.



NEXT

Every insured station, every season.

Today

One screen split into what needs the officer, what the agent is working on, and what closed overnight.

New claims picked up on arrival

The agent opens the case as the claim lands, with the watch case already attached.

Every station on the map

The watcher across all of a cooperative's gauges, with one outbox for every letter and notice.


pluvia
The agent works the claim. You sign.
pluvia.namite.xyz · github.com/jadonamite/pluvia